CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 7: **Particulate Nature of Matter** | Quick revision, one page

Particulate model

- Matter is made of tiny particles; there are spaces between them.
- Particles are always in motion; they attract each other.
- Evidence: dissolving sugar or salt in water, diffusion of ink, smell of perfume spreading.

States of matter

- Solid: particles close, fixed positions (vibrate); definite shape and volume.
- Liquid: particles close but can slide; definite volume, no fixed shape.
- Gas: particles far apart, move freely; no definite shape or volume; easily compressed.

Changes of state

- Melting: solid to liquid (melting point of ice 0 degrees C). Freezing: liquid to solid.
- Boiling/evaporation: liquid to gas (boiling point of water 100 degrees C). Condensation: gas to liquid.
- During a change of state the temperature stays constant; heat is used to change the arrangement of particles.
- Evaporation takes place at all temperatures from the surface and causes cooling.

Rate of evaporation and diffusion

- Evaporation increases with temperature, surface area and wind; decreases with humidity.
- Diffusion is fastest in gases, then liquids, then solids; it increases with temperature.

MASTERMIND